



Course Syllabus

IDEA100 · Technology Rotation

DON BOSCO TECHNICAL INSTITUTE | TECHNOLOGY DEPARTMENT | FALL 2026
MR. PINA, apina@boscotech.edu | HOURS: 7:30, LUNCH, UNTIL 4:30

Course Freshman technology rotation on the daily technology block. One rotation runs about six weeks. Letter grade, not GPA-weighted. This syllabus covers all three rotations.

Enrollment Grade 9. No prerequisite.

The Course

IDEA is engineering design and manufacturing. Over six weeks you will run a mill, a lathe, a bandsaw, and a belt sander, cut and engrave your own dogtag, design a part that gets 3D printed and then loaded until it breaks, and build an IDEA-Blade with a team to fight in a tournament. You take home the hex core, your dogtag, and your hook. The IDEA-Blade is a team build, not an individual take-home.

This is a sampler. At the end of the rotation you move to a different technology, and at the end of freshman year you choose one pathway for the next four.

The Rotation

P1	Manufacturing and Assembly	Manufacture a hex core across five stations, then bench assembly.
P2	Dogtag	Design in CAD, cut and engraved live on the xTool MetalFab.
P3	Hook Competition	Design a hook under a 60g cap, print it, load it to failure.
P4	IDEA-Blade	Teams build a combat spinning top, then a double-elimination tournament.

ROTATION	FIRST DAY	SIGNED PAGE DUE	LAST DAY
Rotation 1	Mon, Aug 17	Fri, Aug 21	Wed, Sep 23
Rotation 2	Thu, Sep 24	Fri, Oct 2	Wed, Oct 28
Rotation 3	Thu, Oct 29	Fri, Nov 6	Mon, Dec 14

ideabosco.com

Everything in IDEA runs through ideabosco.com. Students sign in with their boscotech.net Google account.

IDEA CLASSROOM

ideabosco.com/classroom

Assignments are posted, worked in, and submitted here. Due dates, returned grades, and comments live here too. FACTS remains the gradebook.

THE NOTEBOOK

ideabosco.com/notebook

The engineering record for the rotation. A bound physical notebook is required. Shop work gets written on paper, then photographed and uploaded here.

Grading

CATEGORY	WEIGHT	POINTS
Manufacturing and Assembly	12%	60
Dogtag	8%	40
Hook Competition	26%	130
IDEA-Blade (team 120 + individual 70)	38%	190
Engineering Documentation	16%	80
Rotation total	100%	500

Large projects are staged into checkpoints, so a mistake costs the day it happens instead of the whole project. There is no final exam. Letter grades follow the school's scale. This course is not GPA-weighted. Project due dates, the hook competition event, and the Blade tournament fall on different calendar days in each rotation, and all of them are posted in IDEA Classroom.

IDEA Coins are not grade points. The class runs a classroom currency students earn, lose, and spend on things like printing and small privileges. It does not convert into grade points and has no effect on a student's grade in this course. The full rules are in the digital syllabus.

Three Rules That Matter

- 1 Deadlines are firm.** No extensions. An excused absence moves a due date by the number of days out, and that is the only adjustment. There is typically no homework, but unfinished classwork gets finished outside of class and turned in on time. The shop is open before school from 7:30, at lunch, and after school until 4:30 for exactly that, along with machine time, printing, and help. Machine access outside of class still requires that the student has been trained and cleared on that machine.
- 2 Defend it or fix it.** A student can be asked to explain any part of what they submitted, in their own words, at any time. Work a student cannot explain is corrected the same day, before it earns credit. That is a correction, not a penalty.
- 3 Use AI honestly.** AI is allowed. Students verify anything it gives them against the real source and must be able to explain everything they submit in their own words. Every assignment carries a level from OFF to OPEN and asks for disclosure. Most consumer AI services require account holders to be 18 or older, so students should check before signing up for anything.

Shop Safety

Students work on real manufacturing equipment in this course. These rules are not negotiable, and violating them removes a student from the shop.

- Eye protection any time anyone nearby is cutting, grinding, or machining.
- Closed-toe shoes. No loose clothing or jewelry. Long hair tied back.
- No machine runs until the student is trained on it and cleared for that session.
- Hearing protection where it is called for. Bluetooth headphones and earbuds do not count.
- Every injury, near miss, and broken tool gets reported immediately. Reporting one is never a problem. Hiding one is.
- Horseplay is removal from the shop for the day. No warning.

Cost and Materials

Nothing in this course is a required purchase. IDEA provides all safety gear, tools, and project materials for the rotation.

BRING EVERY DAY

A bound physical notebook, a pen or pencil, and a device that reaches ideabosco.com. The notebook is not optional. Any work done outside the classroom, in the shop or at a test station, gets recorded on paper while it happens.



ideabosco.com/100

Recommended purchases. IDEA provides basic safety gear and required tools at all times, so nothing below is mandatory. Owning your own is recommended for fit and sanitation.

Scan for the digital syllabus, including full project details, the notebook standard, the complete AI policy, and the IDEA Coin rules. If cost is a problem, see me.

Safety glasses, ANSI Z87.1+	\$8-20	Engineering notebook	\$8-12
Digital caliper, 6 in	\$15-25	Whole list, together	\$31-57



Syllabus Acknowledgment

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DETACH AND RETURN | DUE BY THE DATE FOR YOUR ROTATION

ROTATION 1: FRI, AUG 21 | ROTATION 2: FRI, OCT 2 | ROTATION 3: FRI, NOV 6

We have read the IDEA100 syllabus. We understand:

- How the course is graded, that there is no final exam, and that the course is not GPA-weighted.
- That deadlines are firm and no extensions are given.
- That work a student cannot explain when asked is corrected before it earns credit.
- That AI use is allowed, must be verified and explainable, and must be disclosed.
- That students work on real manufacturing equipment, and that violating a shop safety rule removes a student from the shop.
- That a bound physical notebook is required, and that nothing else in this course is a required purchase.

Student

NAME (PRINT)

ROTATION / BLOCK

SIGNATURE

DATE

Parent or Guardian

NAME (PRINT)

SIGNATURE

EMAIL

PHONE

DATE